StockEvent Lens — Batch 001 Intelligence Dashboard

Articles 001–005 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_01 (Articles 001-005)`

TEST 001
Original — 100% Base
DOJ opens antitrust probe into NVIDIA–Groq nonexclusive licensing deal & talent transfer[cite: 51, 56].

TEST 002
Repeat — 96% Overlap
Restates DOJ inquiry into NVIDIA–Groq licensing deal with zero new facts[cite: 51, 52].

TEST 003
Update — +25% New Info
DOJ sets 30-day response deadline for NVIDIA regarding formal information request[cite: 51, 53].

TEST 004
Conflict — Dispute (Exclusive)
Contradicts license terms: claims competing account granted exclusive rights[cite: 51, 54].

TEST 005
Different — 0% Overlap
Unrelated opening of NVIDIA GPU developer training center in Bangkok starting Group 002[cite: 51, 55].

Technical Analysis: Dense Vector Blindness vs. Temporal Contradiction Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *contractual exclusivity accuracy*. Modifying license terms (e.g., "nonexclusive licensing deal" in test_001 vs "exclusive rights" in test_004) yields extremely high vector similarity (~0.94) because both texts share identical surrounding context tokens ("NVIDIA", "Groq", "DOJ inquiry", "December agreement", "inference technology").

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard contractual license type or exclusivity dispute.
```

1. Dense Embedding Blindness
Dense embeddings project words into high-dimensional space where "nonexclusive" and "exclusive" share near-identical context embeddings. Thus, pure vector cosine distance fails to flag this legal terms conflict.

2. LLM Evaluation Override
To prevent mischaracterizing legal structures in regulatory probes, the LLM reasoning layer performs explicit attribute-value exclusivity verification, overriding high vector similarity when $\Delta F_{contradiction} = 1.0$ is detected.

Article Relationship & Percentage Breakdown (Batch 001)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_001_original`	Original (%5=1)	DOJ antitrust probe, NVIDIA Groq deal, nonexclusive license, Jonathan Ross[cite: 51, 56]	100%	Baseline	Establishes breaking story frame on DOJ antitrust inquiry into NVIDIA–Groq licensing deal[cite: 51, 56].
`test_002_repeat`	Repeat (%5=2)	DOJ examining Nvidia Groq agreement, nonexclusive license, Jonathan Ross move[cite: 51, 52]	96%	0% New Facts	Restates identical probe details, nonexclusive terms, and executive transfer without new facts[cite: 51, 52].
`test_003_update`	Update (%5=3)	**30-day response deadline**, formal information request, procedural step[cite: 51, 53]	85%	+25% Procedural Info	Advances event state with a brand-new procedural fact—the imposition of an explicit 30-day response deadline[cite: 51, 53].
`test_004_conflict`	Conflict (%5=4)	**Exclusive rights claim**, disputes nonexclusive terms, competing account[cite: 51, 54]	94% Match	$\Delta F_{contradiction}$ = 1.0	Directly disputes initial terms, asserting competing account claims December agreement granted exclusive rights[cite: 51, 54].
`test_005_different`	Different (%5=0)	Bangkok training center, GPU programming, parallel computing, October start[cite: 51, 55]	0%	Group 002 Trigger	Zero domain overlap; triggers standalone Group 002 event cluster for Bangkok GPU training center[cite: 51, 55].

Group 001 — Consolidated Event Summary

4 Articles (001–004)

The U.S. Department of Justice is examining NVIDIA's December licensing arrangement with AI chip startup Groq over concerns that its structure may have avoided merger review (Articles 001–002)[cite: 51, 52, 56]. The original reporting described the technology license as nonexclusive and noted that Groq executives Jonathan Ross and Sunny Madra subsequently joined NVIDIA[cite: 51, 56]. Procedural development advanced as investigators gave NVIDIA a 30-day response deadline to a formal information request (Article 003)[cite: 51, 53]. However, the contractual terms are in direct conflict: original reports confirmed a nonexclusive license, whereas a competing account claims that the same December license granted exclusive rights (Article 004 ; $\Delta F_{contradiction}$ = 1.0)[cite: 51, 54]. The investigation remains open, and the supplied articles do not establish whether the arrangement violated antitrust law[cite: 51].

✓ Update Preserved: 30-Day Response Deadline from Article 003
✓ Conflict Preserved: Nonexclusive License (001/002) vs Exclusive License (004)
✓ Duplicate Removed: Article 002 Consolidated

Group 002 — Standalone Summary

1 Article (005)

NVIDIA has opened a developer training center in Bangkok focused on GPU programming, parallel computing, and application optimization (Article 005)[cite: 51, 55]. The center plans to begin its first courses in October, offering practical exercises and completion certificates[cite: 51, 55]. Enrollment instructions and class schedules are expected to be published before teaching begins[cite: 51, 55].

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 001

Batch 001 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

test_001_original.json **Group 001**


```
{
  "id": 61707095,
  "headline": "No Acquisition, Same Prize? NVIDIA’s Gr
  "symbols": ["FFOG", "ILDR", "NVDA", "USNZ"],
  "summary": "NVIDIA stock edges lower as the DOJ laun
}
```


Establishes Groq probe & nonexclusive license.

test_003_update.json **Group 001**


```
{
  "id": 990000003,
  "headline": "DOJ Sets 30-Day Response Deadline In Nv
  "symbols": ["FFOG", "ILDR", "NVDA", "USNZ"],
  "summary": "Investigators have given Nvidia 30 days
}
```


Adds 30-day response deadline.

test_004_conflict.json **Group 001**


```
{
  "id": 990000004,
  "headline": "Competing Account Says Nvidia's Decembe
  "symbols": ["FFOG", "ILDR", "NVDA", "USNZ"],
  "summary": "A competing account disputes the origina
}
```


Disputes nonexclusive vs exclusive terms.